PALINDROMIC PERFECT POWERS: AN IMPROVEMENT OF THE CILLERUELO–LUCA–SHPARLINSKI BOUND

ABSTRACT. For an integer base $g \geq 2$ let $\mathcal{P}_{g,L}$ be the set of palindromes with L digits in base g. Cilleruelo, Luca and Shparlinski proved that for every $k \geq 2$ the number of perfect k-th powers in $\mathcal{P}_{g,L}$ is $O\big((\#\mathcal{P}_{g,L})^{1/k}\big)$. We show that the exponent $1/k$ can be replaced by γ/k with an explicit $\gamma = \gamma(g,k) < 1$, for every base g and every k. For instance, one may take $\gamma = 0.9674$ for $g = 2$ and every k, and the number of L-digit palindromic squares in base 10 is $O\big((\#\mathcal{P}_{10,L})^{0.4959}\big)$.

The proof combines the tree argument of Cilleruelo, Luca and Shparlinski with a Fourier analysis of the next blocks of digits from both ends. This reduces the problem to the supremum norm of the Fourier transform of digit reversal, summed over one frequency, together with an elementary equidistribution of archimedean phases across the tree. A power saving for that norm comes from an identity between consecutive digit factors, an idea that goes back to Col. We obtain good explicit constants from a certified computation of suprema over windows of consecutive digits.

1. INTRODUCTION

1.1. **Background.** Let $g \geq 2$ be an integer. Every positive integer n has a unique expansion $n = \sum_{i<L} d_i g^i$ with digits $0 \leq d_i < g$ and $d_{L-1} \neq 0$; we call L the length of n. The integer n is a *palindrome* (in base g) if $d_i = d_{L-1-i}$ for all i. Let $\mathcal{P}_{g,L}$ be the set of palindromes of length L. A palindrome is determined by its first $\lceil L/2 \rceil$ digits, so

$$\#\mathcal{P}_{g,L} = (g-1)g^{\lceil L/2 \rceil - 1} \asymp_g g^{L/2}.$$

Palindromes are a sparse set, and their arithmetic properties have attracted much attention; see for instance [1, 6, 4, 13, 7] and the references there.

Cilleruelo, Luca and Shparlinski [5] studied perfect powers among palindromes. For $k \geq 2$ let

$$\mathcal{Q}^k_{g,L} = \{n \in \mathcal{P}_{g,L} : \ n = x^k \text{ for some } x \in \mathbb{N}\}.$$

They proved

$$\#\mathcal{Q}^k_{g,L} \ \ll_{g,k} \ (\#\mathcal{P}_{g,L})^{1/k}. \tag{1.1}$$

Their argument is elementary. If x^k is a palindrome, its last t digits determine its first t digits. The first t digits confine x to a short interval, and the last t digits confine x to a few residue classes modulo g^t. Write $X_1 = g^{L/k}$, so that $x < X_1$, and note that $X_1 \asymp_{g,k} (\#\mathcal{P}_{g,L})^{2/k}$. Each pair (interval, residue class) contains at most $X_1 g^{-2t} + 1$ admissible x, roughly, and there are about g^t pairs. The choice $g^t \asymp X_1^{1/2}$ gives (1.1). We are not aware of any earlier improvement of the exponent $1/k$ in (1.1), for any base.

1.2. **Results.** Our main result is that (1.1) can be improved by a power, in every base. Let $e(x) = e^{2\pi i x}$ and $F_g(\theta) = \sum_{d=0}^{g-1} e(d\theta)$. For $j \geq 1$ and $0 \leq n < g^j$ let $\mathrm{rev}_j(n)$ be the integer

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obtained by reversing the string of the j digits of n, leading zeros included. For $s \geq 1$ and $\ell, a \in \mathbb{Z}$ put

$$\Psi_\ell(a) = \sum_{m=0}^{g^s-1} e\Big(-\frac{\ell \operatorname{rev}_s(m) + am}{g^s}\Big), \qquad S_\infty(s) = \sum_{0<|\ell|<g^s} \max_{a\in\mathbb{Z}} |\Psi_\ell(a)|.$$

Here $\Psi_\ell(a)$ is the Fourier transform of the function $m \mapsto e(-\ell \operatorname{rev}_s(m)/g^s)$ on $\mathbb{Z}/g^s\mathbb{Z}$. Trivially $S_\infty(s) \leq 2g^{2s}$. Let $\omega(g)$ be the number of distinct prime factors of g.

Theorem 1.1. *Let $g \geq 2$ and $k \geq 2$, and let $0 < c < 1$ be such that $S_\infty(s) \ll_g g^{(2-c)s}$ for all $s \geq 1$. Then the following hold for all $L \geq 3$.*

(a) *If g is a prime power, then*

$$\#\mathcal{Q}_{g,L}^k \ \ll_{g,k,c} \ (\#\mathcal{P}_{g,L})^{\frac{1}{k}\left(1-\frac{c}{4-c}\right)} \log \#\mathcal{P}_{g,L}.$$

(b) *If $c \leq \frac{\log 2}{2\log g}$, then*

$$\#\mathcal{Q}_{g,L}^k \ \ll_{g,k,c} \ (\#\mathcal{P}_{g,L})^{\frac{1}{k}\left(1-\frac{c}{(k-1)(4-c)+2}\right)} \big(\log \#\mathcal{P}_{g,L}\big)^{\omega(g)+1}.$$

If $S_\infty(s) \ll g^{(2-c)s}$ then the same holds for every smaller c, so the condition in (b) is harmless. Part (b) applies to every base; part (a) is stronger for prime powers. The difference comes from integers x that share a large factor with a power of g; this matters only when g is not a prime power (see Remark 6.1).

Theorem 1.2. *For every $g \geq 2$ there is $c_g > 0$ with $S_\infty(s) \ll_g g^{(2-c_g)s}$ for all $s \geq 1$. For the bases in Table 1 one may take the values of c_g listed there.*

TABLE 1. Admissible values of c_g in Theorem 1.2 (rounded down).

g	2	3	4	5	6	7	8	9	10	12	16
c_g	0.1263	0.1002	0.0811	0.0709	0.0641	0.0597	0.0543	0.0516	0.0496	0.0458	0.0383

The existence of some $c_g > 0$ is proved analytically in Section 5.2. The values in Table 1 come from a finite computation, described in Sections 5.3 and 5.4, whose output is certified by a Collatz–Wielandt bound. It uses 40-digit arithmetic where this matters and double precision elsewhere, with a safety margin that dominates the rounding errors; the accompanying script reproduces it. Combining the two theorems (and absorbing the logarithms by rounding) gives the following.

Corollary 1.3. *For every $g \geq 2$ and $k \geq 2$ there is $\gamma = \gamma(g,k) < 1$ with $\#\mathcal{Q}_{g,L}^k \ll_{g,k} (\#\mathcal{P}_{g,L})^{\gamma/k}$. For the bases below one may take the following values.*

prime power g (every k)	2	3	4	5	7	8	9	16
γ	0.9674	0.9744	0.9794	0.9820	0.9849	0.9863	0.9870	0.9904

other g	6	10	12
γ, $k=2$	0.9893	0.9917	0.9924
γ, $k=3$	0.9936	0.9950	0.9954

In particular, the number of L-digit palindromic squares in base 10 is $\ll (\#\mathcal{P}_{10,L})^{0.4959}$.

For prime powers the table lists $1-c_g/(4-c_g)$, and for the other bases $1-c_g/((k-1)(4-c_g)+2)$, rounded up. If g is not a prime power, the saving decays like $1/k$ as k grows.

1.3. **Outline of the method.** We start from the tree of [5]: fix t, and for each residue class b modulo g^t (a *node*) the admissible x form an arithmetic progression $x = x_b + g^t z$, $0 \le z < Z_b$, of length about $X_1 g^{-2t}$. The argument of [5] only uses the first and last t digits. We also use the next s digits from both ends. Inside a node, the digits of x^k in positions $t, \dots, t+s-1$ are an affine function of z modulo g^s (a g-adic linearisation), and the digits in positions $L-t-s, \dots, L-t-1$ are, up to a small error, an affine function of z over $\mathbb{R}$ (an archimedean linearisation). The palindrome condition says that the second block is the reversal of the first. Detecting it with a Beurling–Selberg majorant and expanding $e(-\ell \operatorname{rev}_s(n)/g^s)$ in additive characters reduces the count to geometric series in z with phases $\theta_{b,\ell,a} = (\ell\delta_b + a\beta_b)/g^s$, weighted by $|\Psi_\ell(a)|$ (Proposition 3.4). Here δ_b comes from the size of x and $\beta_b = kb^{k-1} \bmod g^s$ from its residue.

We then bound the weights by $\max_a |\Psi_\ell(a)|$, and recover the averaging that is needed from the archimedean phases $\ell\delta_b$ alone: since δ_b is a monotone function of the first t digits of x^k, and different nodes have essentially different first digits, these phases are well distributed modulo 1 across the nodes (Lemma 4.3). This reduces everything to $S_\infty(s)$.

A power saving for $S_\infty(s)$ comes from an identity between consecutive factors of the product formula for $\Psi_\ell(a)$, in which the frequency a cancels (Lemma 5.1): a large value of $|\Psi_\ell(a)|$ needs all factors to be large, and the identity makes this impossible for most ℓ, whatever a is. Arguments of this kind go back to Col [6]; see also [13], [7, Section 2.3] and [2]. We give a self-contained treatment, and we sharpen the constants considerably by observing that the identity determines the whole sequence of factor arguments from the first one (Lemma 5.2). This allows us to bound the supremum over a on windows of consecutive digits, and to sum over ℓ with a transfer matrix (Proposition 5.7).

If g is not a prime power, there is one more difficulty: an integer x may be divisible by a high power of one prime factor of g, and for such nodes the reduction loses. We treat these x separately, by a direct count (Lemma 6.2).

1.4. **Related work.** Uniform bounds for exponential sums involving digit reversal appear in Col [6], Tuxanidy and Panario [13], Dartyge, Martin, Rivat, Shparlinski and Swaenepoel [7] (in base 2) and Bhowmik and Suzuki [2] (in general bases, in a preprint). In base 2, the mechanism behind Lemma 5.1 appears in [7, Lemmas 2.5 and 2.6], where it is credited to an idea of Col [6, proof of Lemme 2]. In general bases, the identity of Lemma 5.1, in the form dual to ours under Lemma 2.1(iv), appears in the proof of [2, Lemma 4.6], a lemma which Bhowmik and Suzuki attribute to Tuxanidy and Panario [13]; they combine it with a pairing of consecutive factors [2, Lemmas 4.7 and 4.9], as we do in Lemma 5.3. We do not use any result of [2]: everything we need is proved here. Johnston and Kerr [10] (also a preprint) combine the argument of [5] with a hybrid p-adic/archimedean van der Corput method to prove that every base has infinitely many square-free palindromes. They count palindromes $n \le x$ coprime to $g^3 - g$ and divisible by d^2 for some $d \in [D, 2D]$; their bounds that go beyond the elementary method of [5] require $D \le x^{2/5}$, while their elementary bound, for $D \asymp x^{1/2}$, recovers (1.1) for squares. Square and cube divisors of palindromes also play a role in the recent preprints of Chourasiya and Johnston [3] and of Tuxanidy [12]. We do not use the results of these preprints.

1.5. **Notation.** We use Vinogradov's notation $\ll$ and $\gg$, and $A \asymp B$ for $A \ll B \ll A$; implied constants may depend on g and k, and on other parameters only when indicated. We write $\|x\|$ for the distance from x to the nearest integer, v_p for the p-adic valuation, and $0.d_1 d_2 \cdots d_r$ for $\sum_j d_j g^{-j}$. Throughout, $X_1 = g^{L/k}$; then x^k has length L if and only if $g^{(L-1)/k} \le x < X_1$, and $X_1 \asymp (\#\mathcal{P}_{g,L})^{2/k}$. Finally

$$f(\theta) = \frac{|F_g(\theta)|}{g} = \left|\frac{\sin \pi g\theta}{g \sin \pi\theta}\right| \quad (\theta \notin \mathbb{Z}), \qquad f(\theta) = 1 \quad (\theta \in \mathbb{Z}).$$

Thus $0 \le f \le 1$, f is even and 1-periodic, $f(\theta) < 1$ for $\theta \notin \mathbb{Z}$, and f is continuous.

2. Preliminaries

Lemma 2.1. *Let $s \ge 1$ and $\ell, a \in \mathbb{Z}$.*

(i) *For $0 \le n < g^s$, $e(-\ell \operatorname{rev}_s(n)/g^s) = g^{-s} \sum_{a \bmod g^s} \Psi_\ell(a) e(an/g^s)$.*
(ii) *$|\Psi_\ell(a)| = \prod_{i=0}^{s-1} |F_g(\theta_i)| = g^s \prod_{i=0}^{s-1} f(\theta_i)$, where $\theta_i = \theta_i(\ell, a) = (\ell g^{s-1-i} + a g^i)/g^s$.*
(iii) *If $\ell \not\equiv 0 \pmod{g^s}$ then $\Psi_\ell(0) = 0$.*
(iv) *$|\Psi_\ell(a)| = |\Psi_a(\ell)|$.*

Proof. (i) is Fourier inversion on $\mathbb{Z}/g^s\mathbb{Z}$. (ii) Write $m = \sum_{i<s} m_i g^i$; then $\operatorname{rev}_s(m) = \sum_i m_i g^{s-1-i}$, and $\Psi_\ell(a) = \prod_{i<s} F_g(-\theta_i)$; note $|F_g(-\theta)| = |F_g(\theta)|$. (iii) Since rev_s permutes $[0, g^s)$, $\Psi_\ell(0) = \sum_{n<g^s} e(-\ell n/g^s) = 0$. (iv) Substitute $m = \operatorname{rev}_s(m')$ in the definition; rev_s is an involution of $[0, g^s)$. □

Lemma 2.2 (Beurling–Selberg majorant)**.** *Let $J \subset \mathbb{R}/\mathbb{Z}$ be an arc of length $|J| < 1$ and let $\kappa \ge 0$ be an integer. There is a real trigonometric polynomial $S(x) = \sum_{|\ell| \le \kappa} \hat{S}(\ell) e(\ell x)$ with $S \ge \mathbf{1}_J$ and $\hat{S}(0) = |J| + 1/(\kappa+1)$. Moreover $|\hat{S}(\ell)| \le \hat{S}(0)$ for all ℓ.*

Proof. The first statement is Selberg's construction; see [14] or [11, Chapter 1]. Since $S \ge 0$, $|\hat{S}(\ell)| \le \int_0^1 S = \hat{S}(0)$. □

Lemma 2.3. *For $\theta \in \mathbb{R}$ and an integer $Z \ge 1$, $\left|\sum_{z=0}^{Z-1} e(z\theta)\right| \le \min\left(Z, \frac{1}{2\|\theta\|}\right)$.*

Proof. For $\theta \notin \mathbb{Z}$ the sum is $(e(Z\theta)-1)/(e(\theta)-1)$, of modulus at most $1/|\sin \pi\theta| \le 1/(2\|\theta\|)$. □

Lemma 2.4. *For every integer $n \ge 1$, the group $(\mathbb{Z}/n\mathbb{Z})^\times$ has at most $(2k)^{\omega(n)}$ elements u with $u^k = 1$.*

Proof. By the Chinese remainder theorem it suffices to treat $n = p^a$. For odd p, or $p^a \le 4$, the group $(\mathbb{Z}/p^a\mathbb{Z})^\times$ is cyclic and has at most k such elements; for $a \ge 3$, $(\mathbb{Z}/2^a\mathbb{Z})^\times \cong \mathbb{Z}/2 \times \mathbb{Z}/2^{a-2}$ has at most $2k$ (see [9, Chapter 4]). □

3. The tree and the reduction

3.1. **Nodes.** Fix integers $t \ge 3$ and $1 \le s \le t-2$ with $t + s \le L/2$. A *node* is an integer $b \in [0, g^t)$ with $g \nmid b^k$. For a node b put

$$T_b = \operatorname{rev}_t(b^k \bmod g^t), \qquad I_b = \left[(T_b g^{L-t})^{1/k},\ ((T_b+1)g^{L-t})^{1/k}\right),$$

and let Z_b be the number of integers $x \in I_b$ with $x \equiv b \pmod{g^t}$. If $Z_b \ge 1$ let x_b be the least of them, and put

$$\beta_b = k b^{k-1} \bmod g^s, \qquad \delta_b = k x_b^{k-1} g^{2t+s-L}, \qquad \theta_{b,\ell,a} = \frac{\ell\delta_b + a\beta_b}{g^s}.$$

Since $g \nmid b^k$, the last digit of b^k is non-zero, so $g^{t-1} \le T_b < g^t$. Finally put

$$Z = X_1 g^{-2t}, \qquad \delta(T) = k(T g^{L-t})^{(k-1)/k} g^{2t+s-L} \quad (g^{t-1} \le T < g^t).$$

Lemma 3.1 ([5])**.** *If $x^k \in \mathcal{P}_{g,L}$, then $b = x \bmod g^t$ is a node, $x \in I_b$, and $x = x_b + g^t z$ for some $0 \le z < Z_b$.*

Proof. The last digit of a palindrome equals its first digit, which is non-zero; so $g \nmid x^k$, and hence $g \nmid b^k$. The number formed by the first t digits of x^k is $\lfloor x^k / g^{L-t} \rfloor$, and by the palindrome property it equals $\operatorname{rev}_t(x^k \bmod g^t) = \operatorname{rev}_t(b^k \bmod g^t) = T_b$. Hence $T_b g^{L-t} \le x^k < (T_b+1)g^{L-t}$, i.e. $x \in I_b$. □

Lemma 3.2. *For every node, $|I_b|g^{-t} \in [Z/k,\ gZ/k]$, so $Z/k-1 \le Z_b \le gZ/k+1$; in particular $Z_b \ge 1$ if $Z \ge 2k$. If $Z_b \ge 1$ then $0 \le \delta_b - \delta(T_b) \le g^{1-t}\delta(T_b)$. Finally $\frac{k}{2g}\,\frac{g^s}{Z} \le \delta(T) \le k\,\frac{g^s}{Z}$ for $g^{t-1} \le T < g^t$.*

Proof. Let $g^{t-1} \le T < g^t$, $y_0 = Tg^{L-t}$ and $y_1 = y_0 + g^{L-t}$. By the mean value theorem $y_1^{1/k} - y_0^{1/k} = g^{L-t}/(k\xi^{(k-1)/k})$ for some $\xi \in (y_0, y_1) \subset [g^{L-1}, g^L]$, which gives the first claim (take $T = T_b$). Next, $x_b \in [y_0^{1/k}, y_1^{1/k})$ for $T = T_b$, so $\delta_b/\delta(T_b) = (x_b/y_0^{1/k})^{k-1} \in [1, (y_1/y_0)^{(k-1)/k}]$, and $y_1/y_0 = 1 + 1/T_b \le 1 + g^{1-t}$. Finally $\delta(T)\cdot g^{L-t}/(k\xi^{(k-1)/k}) = g^{t+s}(y_0/\xi)^{(k-1)/k} \in [g^{t+s}/2, g^{t+s}]$, because $y_0/\xi \ge y_0/y_1 \ge \frac12$; combine with $g^{L-t}/(k\xi^{(k-1)/k}) \in [g^tZ/k, g^{t+1}Z/k]$. □

3.2. **Linearisation.** For a node b with $Z_b \ge 1$ put $\alpha_b = \lfloor x_b^k/g^t \rfloor \bmod g^s$, and for real $z \ge 0$

$$\varphi_b(z) = \frac{(x_b + g^t z)^k}{g^{L-t-s}} - T_b g^s, \qquad \lambda_b(z) = \varphi_b(0) + \delta_b z.$$

Lemma 3.3. *Let b be a node and $x = x_b + g^t z$ with $0 \le z < Z_b$.*

(a) *The digits of x^k in positions $t, \dots, t+s-1$ form the number $n_b(z) := (\alpha_b + \beta_b z) \bmod g^s$.*
(b) *The digits of x^k in positions $L-t-s, \dots, L-t-1$ form the number $\lfloor \varphi_b(z) \rfloor$, and $0 \le \varphi_b(z) - \lambda_b(z) \le \frac{k-1}{k} g^{s+1-t} \le \frac{1}{g}$.*
(c) *If $x^k \in \mathcal{P}_{g,L}$, then $\lambda_b(z) - \mathrm{rev}_s(n_b(z)) \in [-\frac1g, 1)$.*

Proof. (a) Write $x^k = x_b^k + kx_b^{k-1}g^t z + g^{2t}N'$ with $N' \in \mathbb{Z}$. Then $\lfloor x^k/g^t \rfloor = \lfloor x_b^k/g^t \rfloor + kx_b^{k-1}z + g^tN'$; reduce modulo g^s, using $s \le t$ and $x_b \equiv b \pmod{g^t}$.

(b) For $x \in I_b$ we have $\lfloor x^k/g^{L-t} \rfloor = T_b$, so the block in question is $\lfloor x^k/g^{L-t-s} \rfloor - T_b g^s = \lfloor \varphi_b(z) \rfloor$. As a function of real z, φ_b is convex with $\varphi_b'(0) = \delta_b$, so $\varphi_b \ge \lambda_b$, and by Taylor's theorem

$$\varphi_b(z) - \lambda_b(z) \le \tfrac12 k(k-1)\, x_{\max}^{k-2} (g^t z)^2 g^{-(L-t-s)},$$

where $x_{\max} = \sup I_b$. With y_0, y_1 as in the proof of Lemma 3.2 (for $T = T_b$) we have $g^t z \le |I_b| \le g^{L-t}/(ky_0^{1-1/k})$ and $x_{\max} \le y_1^{1/k}$, so

$$x_{\max}^{k-2}(g^t z)^2 \le \Big(\frac{y_1}{y_0}\Big)^{\frac{k-2}{k}} \frac{g^{2(L-t)}}{k^2 y_0} \le \frac{2g^{L-2t+1}}{k^2},$$

using $y_0 \ge g^{L-1}$ and $y_1 \le 2y_0$. This gives the bound $\frac{k-1}{k} g^{s+1-t}$, which is at most $1/g$ because $s \le t-2$.

(c) Since $t + s \le L/2$, the two blocks are disjoint, and the palindrome property says that the digit in position $L-1-j$ equals the digit in position j. Hence the number formed by the block in (b) is the reversal of the number formed by the block in (a): $\lfloor \varphi_b(z) \rfloor = \mathrm{rev}_s(n_b(z))$. So $\mathrm{rev}_s(n_b(z)) \in (\varphi_b(z) - 1, \varphi_b(z)]$, and the claim follows from (b). □

3.3. **The reduction.** For a set $\mathcal{B}$ of nodes with $Z_b \ge 1$ for all $b \in \mathcal{B}$, put

$$E(\mathcal{B}) = \sum_{b \in \mathcal{B}} \sum_{0 < |\ell| < g^s} \sum_{a \not\equiv 0 \ (g^s)} |\Psi_\ell(a)| \min\Big(Z_b, \frac{1}{2\|\theta_{b,\ell,a}\|}\Big).$$

Proposition 3.4. *For every such $\mathcal{B}$,*

$$\#\{x :\ x^k \in \mathcal{P}_{g,L},\ x \bmod g^t \in \mathcal{B}\} \ \le\ \frac52 g^{-s} \sum_{b \in \mathcal{B}} Z_b + \frac52 g^{-2s} E(\mathcal{B}).$$

Proof. Let J be the image of $[-\frac{1}{2},1]$ in $\mathbb{R}/g^s\mathbb{Z}$. Applying Lemma 2.2 to the arc $g^{-s}J$ of $\mathbb{R}/\mathbb{Z}$ with $\kappa = g^s-1$ and rescaling, we get $F(y)=\sum_{|\ell|\le\kappa} c_\ell e(\ell y/g^s)$ with $F\ge \mathbf{1}_J$ on $\mathbb{R}$, $c_0=\frac{3/2}{g^s}+\frac{1}{g^s}=\frac{5}{2g^s}$ and $|c_\ell|\le c_0$.

Fix $b\in\mathcal{B}$. By Lemmas 3.1 and 3.3(c), the number of solutions x with $x\equiv b \pmod{g^t}$ is at most

$$\sum_{z<Z_b} F\big(\lambda_b(z)-\mathrm{rev}_s(n_b(z))\big)=\sum_{|\ell|\le\kappa} c_\ell \sum_{z<Z_b} e\Big(\frac{\ell\lambda_b(z)}{g^s}\Big)e\Big(-\frac{\ell\,\mathrm{rev}_s(n_b(z))}{g^s}\Big).$$

Expand the last factor by Lemma 2.1(i). Since $e(an_b(z)/g^s)=e(a(\alpha_b+\beta_b z)/g^s)$ and $\lambda_b(z)=\lambda_b(0)+\delta_b z$, the sum over z becomes $g^{-s}\sum_a \Psi_\ell(a)\,e(\cdot)\sum_{z<Z_b}e(z\theta_{b,\ell,a})$, with a unimodular factor $e(\cdot)$ independent of z. By Lemma 2.3 its modulus is at most $g^{-s}\sum_a|\Psi_\ell(a)|\min(Z_b,1/(2\|\theta_{b,\ell,a}\|))$. For $\ell=0$, $\Psi_0(a)=g^s\mathbf{1}_{a\equiv0}$ and the term is c_0Z_b. For $0<|\ell|\le\kappa$ we have $\ell\not\equiv0 \pmod{g^s}$, so the term $a\equiv0$ vanishes by Lemma 2.1(iii). Sum over $b\in\mathcal{B}$ and use $|c_\ell|\le c_0$. □

4. The supremum over the g-adic frequency

We now bound the weights $|\Psi_\ell(a)|$ in $E(\mathcal{B})$ by their maximum over a, sum the remaining factor over a (Lemma 4.1), and then over the nodes (Lemma 4.3).

Lemma 4.1. *Let $y\in\mathbb{R}$, $\beta\in\mathbb{Z}$, $d=\gcd(\beta,g^s)$ and $A\ge1$. Then*

$$\sum_{a \bmod g^s}\min\Big(A,\frac{1}{2\|(y+a\beta)/g^s\|}\Big)\ \le\ \min\Big(dA,\frac{g^s}{2\|y/d\|}\Big)+g^s(2+s\log g).$$

Proof. As a runs modulo g^s, $a\beta$ runs over the multiples of d modulo g^s, each taken d times. Put $Q=g^s/d$ and $y/d=n_0+\epsilon_0$ with $n_0\in\mathbb{Z}$, $|\epsilon_0|=\|y/d\|$. Then $(y+cd)/g^s=(n_0+c+\epsilon_0)/Q$, and as c runs modulo Q the residue $j\equiv n_0+c$ runs over $(-Q/2,Q/2]$. For $j=0$ the distance to $\mathbb{Z}$ is $|\epsilon_0|/Q$; for $j\ne0$ it is at least $(|j|-\frac12)/Q$. Hence the sum is at most $d\min(A,Q/(2|\epsilon_0|))+d\sum_{j\ne0}Q/(2|j|-1)\le\min(dA,g^s/(2|\epsilon_0|))+g^s(2+\log Q)$. □

For a non-zero integer n let $m_g(n)=\prod_{p\mid g}p^{v_p(n)}$ be its *g-part*, so that $n/m_g(n)$ is coprime to g. For $D\ge1$ let $\tau_g(D)=\#\{m\le D:\ m\mid g^\infty\}\le(1+\log_2 D)^{\omega(g)}$.

Lemma 4.2 (nodes with small g-part)**.** *Let $D\ge1$ with $D^k<2^t$, and let b be a node with $m:=m_g(b)\le D$.*

(i) *$d_b:=\gcd(\beta_b,g^s)$ divides km^{k-1}.*
(ii) *Every node b' with $T_{b'}=T_b$ has $m_g(b')=m$, and there are at most $(2k)^{\omega(g)}m^{k-1}$ such nodes.*

If $g=p^e$ is a prime power, every node has $m_g(b)<g^{1/k}$.

Proof. Write $b=mu$ with $\gcd(u,g)=1$. (i) $\beta_b\equiv km^{k-1}u^{k-1} \pmod{g^s}$, so $d_b=\gcd(km^{k-1},g^s)$. (ii) Let $p\mid g$, $e_p=v_p(g)$ and $v=v_p(b)$. Then $p^{kv}\le m^k<2^t\le p^{e_pt}$, so $kv<e_pt$. If $T_{b'}=T_b$ then $b'^k\equiv b^k \pmod{p^{e_pt}}$, and the right side has p-adic valuation exactly $kv<e_pt$; hence $v_p(b')=v$. So $m_g(b')=m$, and $b'=mu'$ with $\gcd(u',g)=1$ and $u'^k\equiv u^k \pmod{g^t/m^k}$ (note $m^k\mid g^t$). By Lemma 2.4, u' lies in at most $(2k)^{\omega(g)}$ classes modulo g^t/m^k. Since $b' \bmod g^t$ is determined by $u' \bmod g^t/m$, and each class modulo g^t/m^k has m^{k-1} lifts modulo g^t/m, there are at most $(2k)^{\omega(g)}m^{k-1}$ such b'. Finally, if $g=p^e$ then $p^e\nmid b^k$ gives $kv_p(b)<e$. □

Lemma 4.3 (archimedean count)**.** *Let $s\le t-2$, $Z\ge2k$, and $D\ge1$ with $D^k<2^t$. Let $\mathcal{B}$ be a set of nodes with $m_g(b)\le D$ for all $b\in\mathcal{B}$, and for $0<|\ell|<g^s$ put*

$$W_\ell(\mathcal{B})=\sum_{b\in\mathcal{B}}\sum_{a\not\equiv0}\min\Big(Z_b,\frac{1}{2\|\theta_{b,\ell,a}\|}\Big).$$

Then, with $\tau = \tau_g(kD^{k-1})$,

$$W_\ell(\mathcal{B}) \ll \tau D^{k-1}\Big((\log DZ)\, g^{t+s} + g^{2s} + D^{k-1}Z\Big) + \tau D^{k-2}\,\frac{Zg^t}{|\ell|} + (s+1)\, g^{t+s}.$$

Proof. By Lemma 3.2, $Z_b \geq 1$ for every node. By Lemma 4.1 with $A = Z_b \leq Z_+ := gZ/k + 1$,

$$W_\ell(\mathcal{B}) \leq \sum_{b\in\mathcal{B}} \min\Big(d_b Z_+, \frac{g^s}{2\|\ell\delta_b/d_b\|}\Big) + g^{t+s}(2 + s\log g).$$

Split $\mathcal{B}$ into $\mathcal{B}_1 = \{b : |\ell|\delta_b \leq d_b/2\}$ and $\mathcal{B}_2 = \mathcal{B} \setminus \mathcal{B}_1$.

Small phases. For $b \in \mathcal{B}_1$ we have $\|\ell\delta_b/d_b\| = |\ell|\delta_b/d_b$, and $\delta_b \geq \delta(T_b) \geq kg^s/(2gZ)$ by Lemma 3.2. So the term for b is at most $gZd_b/(k|\ell|)$. By Lemma 4.2(i), and since at most g^t/m residues modulo g^t are divisible by a given $m \mid g^t$,

$$\sum_{b\in\mathcal{B}} d_b \leq \sum_{\substack{m\leq D\\ m\mid g^\infty}} \frac{g^t}{m}\, km^{k-1} \leq k\,\tau_g(D)\, D^{k-2} g^t.$$

This gives the term $\tau D^{k-2} Zg^t/|\ell|$, since $\tau_g(D) \leq \tau$.

Large phases: reduction to a count over T. Let $b \in \mathcal{B}_2$. By Lemma 3.2 and $s \leq t-2$, $|\ell\delta_b - \ell\delta(T_b)|/d_b \leq g^s \cdot \frac{kg^s}{Z} \cdot \frac{g^{1-t}}{d_b} \leq \frac{kg^{s-1}}{d_b Z} =: \varepsilon_1$. Replace δ_b by $\delta(T_b)$ in the minimum. If $\|\ell\delta(T_b)/d_b\| \geq 2\varepsilon_1$, the old minimum is at most twice the new one. Otherwise the new minimum is at least $\min(d_bZ_+, g^s/(4\varepsilon_1)) \geq gd_bZ/(4k)$, while the old one is at most $d_bZ_+ \leq gd_bZ/k + d_b$; so again the ratio is at most $4 + 4k/(gZ) \leq 6$. Also $|\ell|\delta(T_b) \geq |\ell|\delta_b/(1+g^{1-t}) > d_b/3$. Group the nodes of $\mathcal{B}_2$ by $d = d_b$; by Lemma 4.2(i) there are at most τ values of d, all at most kD^{k-1}. By Lemma 4.2(ii), each T equals T_b for at most $(2k)^{\omega(g)} D^{k-1}$ nodes $b \in \mathcal{B}$. Hence

$$\sum_{b\in\mathcal{B}_2} \min\Big(d_bZ_+, \frac{g^s}{2\|\ell\delta_b/d_b\|}\Big) \ll D^{k-1} \sum_d \sum_{T\in\mathcal{T}_d} \min\Big(dZ_+, \frac{g^s}{2\|h_d(T)\|}\Big),$$

where $h_d(T) = \ell\delta(T)/d$ and $\mathcal{T}_d = \{T \in [g^{t-1}, g^t) : |h_d(T)| \geq \frac{1}{3}\}$, an interval.

The count. On $[g^{t-1}, g^t)$ the function h_d is monotone with $|h_d'(T)| = \frac{k-1}{k}|h_d(T)|/T$, and $|h_d'|$ varies by a factor at most g there, since $|h_d'| \propto T^{-1/k}$. Let $\mu = \min_{\mathcal{T}_d} |h_d'|$; then $\mu \geq \frac{k-1}{3k} g^{-t}$. The range of h_d on $\mathcal{T}_d$ has length $R \leq g^{t+1}\mu$, and also $R \leq |\ell| \max\delta/d \leq k|\ell|g^s/(dZ)$. For every $u > 0$, the $T \in \mathcal{T}_d$ with $\|h_d(T)\| \leq u$ lie in at most $R+2$ intervals of length at most $2u/\mu$, each containing at most $2u/\mu + 1$ integers, so

$$n(u) := \#\{T \in \mathcal{T}_d : \|h_d(T)\| \leq u\} \leq (R+2)\Big(\frac{2u}{\mu} + 1\Big) \leq 2ug^{t+1} + \frac{4u}{\mu} + R + 2 \ll ug^{t+1} + R + 2.$$

Split the T dyadically according to $\|h_d(T)\|$, starting at H/A' with $A' = dZ_+$ and $H = g^s/2$; there are $j_0 + 2$ ranges with $j_0 \ll \log(DZ)$. This gives

$$\sum_{T\in\mathcal{T}_d} \min\Big(A', \frac{H}{\|h_d(T)\|}\Big) \leq \sum_{j=-1}^{j_0} \frac{A'}{2^j}\, n\Big(\frac{2^{j+1}H}{A'}\Big) \ll (j_0+2)\, g^{t+1}H + A'(R+2).$$

Here $A'R \ll |\ell|g^s \leq g^{2s}$ and $A' \leq kD^{k-1}Z_+$. Summing over the at most τ values of d gives the claim. □

Corollary 4.4. *Under the hypotheses of Lemma 4.3,*

$$E(\mathcal{B}) \;\ll\; \tau D^{k-1}\Big((\log DZ)\, g^{t+s} + g^{2s} + D^{k-1}Z\Big)S_\infty(s) + \tau D^{k-2}(s+1)\, Zg^{t+s} + (s+1)\, g^{t+s} S_\infty(s).$$

Proof. Clearly $E(\mathcal{B}) \le \sum_{0<|\ell|<g^s} \max_a |\Psi_\ell(a)|\, W_\ell(\mathcal{B})$. Insert Lemma 4.3. In the term with $1/|\ell|$, use $\max_a |\Psi_\ell(a)| \le g^s$ and $\sum_{0<|\ell|<g^s} |\ell|^{-1} \le 2(1 + s\log g)$. □

5. The supremum norm of the reversal transform

5.1. **The identity.** Recall from Lemma 2.1(ii) that $|\Psi_\ell(a)| = g^s \prod_{i<s} f(\theta_i)$ with $\theta_i = (\ell g^{s-1-i} + ag^i)/g^s$.

Lemma 5.1. *For all ℓ, a and $0 \le i \le s-2$, $g\theta_i - \theta_{i+1} = (g^2-1)\ell/g^{i+2}$. Consequently*

$$g\|\theta_i\| + \|\theta_{i+1}\| \;\ge\; \Delta_i(\ell) := \Big\| \frac{(g^2-1)\ell}{g^{i+2}} \Big\|.$$

Proof. $g\theta_i = \ell/g^i + a/g^{s-1-i}$ and $\theta_{i+1} = \ell/g^{i+2} + a/g^{s-1-i}$. Subtract. □

The point is that a cancels: a large value of $|\Psi_\ell(a)|$ needs every θ_i to be close to an integer, and this is impossible when $\Delta_i(\ell)$ is not small, whatever a is. In fact the identity determines the whole sequence (θ_i) from θ_0:

Lemma 5.2 (forced orbit)**.** *Let $\ell' = (g^2-1)\ell \bmod g^s$ and, for $0 \le i \le s-2$, $D_i = \ell'/g^{i+2} \bmod 1 = 0.\ell'_{i+1}\ell'_i \cdots \ell'_0$, where ℓ'_j are the digits of ℓ'. Then $\Delta_i(\ell) = \|D_i\|$ and $\theta_{i+1} \equiv g\theta_i - D_i \pmod 1$ for every a. The map $\ell \mapsto \ell'$ is a bijection of $\mathbb{Z}/g^s\mathbb{Z}$.*

Proof. By Lemma 5.1, $\theta_{i+1} = g\theta_i - (g^2-1)\ell/g^{i+2}$, and $(g^2-1)\ell \equiv \ell' \pmod{g^s}$ with $i+2 \le s$. Since $\gcd(g^2-1, g) = 1$, multiplication by g^2-1 is a bijection modulo g^s. □

5.2. **A power saving in every base.** Let $\psi(\epsilon) = \sup\{f(\theta) : \|\theta\| \ge \epsilon\}$, $\phi = -\log\psi \in [0,\infty]$, and

$$\chi(\Delta) = \inf\{\phi(x) + \phi(y) :\; x, y \in [0, \tfrac12],\; gx + y \ge \Delta\}.$$

The function ψ is non-increasing, and ϕ and χ are non-decreasing. Since f is continuous and $f < 1$ off $\mathbb{Z}$, we have $\psi(\epsilon) < 1$, i.e. $\phi(\epsilon) > 0$, for $\epsilon > 0$; and $\chi(\Delta) \ge \phi(\Delta/(g+1)) > 0$ for $\Delta > 0$, because $gx + y \ge \Delta$ forces $\max(x, y) \ge \Delta/(g+1)$.

Lemma 5.3. *For every ℓ, $\max_a |\Psi_\ell(a)| \le g^s \exp\big(-\frac12 \sum_{i=0}^{s-2} \chi(\Delta_i(\ell))\big)$.*

Proof. We have $f(\theta_i) \le \psi(\|\theta_i\|)$, so $\log(|\Psi_\ell(a)|/g^s) \le -\sum_{i<s} \phi(\|\theta_i\|)$. Each index occurs in at most two of the pairs $\{i, i+1\}$, $0 \le i \le s-2$, so $\sum_{i<s} \phi(\|\theta_i\|) \ge \frac12 \sum_{i\le s-2} \big(\phi(\|\theta_i\|) + \phi(\|\theta_{i+1}\|)\big) \ge \frac12 \sum_{i\le s-2} \chi(\Delta_i(\ell))$ by Lemma 5.1. □

Proposition 5.4. *For every $g \ge 2$ there is $c > 0$ with $S_\infty(s) \ll_g g^{(2-c)s}$.*

Proof. By Lemma 5.2 we may sum over ℓ' instead of ℓ; each non-zero residue is represented by two values of ℓ with $0 < |\ell| < g^s$. Let $r = 1$ if $g \ge 3$ and $r = 2$ if $g = 2$. By Lemma 5.2, $\Delta_i(\ell) = \|D_i\|$, and $D_i = 0.\ell'_{i+1}\ell'_i \cdots \ell'_0$ lies in the interval $[v, v + g^{-r}]$ with $v = 0.\ell'_{i+1} \cdots \ell'_{i+2-r}$. Hence $\Delta_i(\ell) \ge \Lambda_r(\ell'_{i+1}, \dots, \ell'_{i+2-r})$, where $\Lambda_r(d_1, \dots, d_r) = \max(0, \min(v, 1 - v - g^{-r}))$ for $v = 0.d_1 \cdots d_r$. By Lemma 5.3 and the monotonicity of χ,

$$S_\infty(s) \le 2g^s \sum_{\ell' \bmod g^s} \prod_{i=0}^{s-2} \xi(\ell'_{i+1}, \dots, \ell'_{i+2-r}), \qquad \xi(d_1, \dots, d_r) = e^{-\chi(\Lambda_r(d_1,\dots,d_r))/2}.$$

If $g \ge 3$ and $r = 1$, the product is over single digits, and the sum is at most $g\rho^{s-1}$ with $\rho = \sum_{d<g} \xi(d)$. Here $\xi(d) < 1$ for $1 \le d \le g-2$, since then $\Lambda_1(d) \ge 1/g > 0$; so $\rho < g$. If $g = 2$ and $r = 2$, then $\Lambda_2(d_1, d_2) = \frac14$ when $d_1 \ne d_2$, so the sum is at most $2\|M^{s-1}\|_\infty$ for the matrix $M = \left(\begin{smallmatrix} 1 & u \\ u & 1 \end{smallmatrix}\right)$ with $u = e^{-\chi(1/4)/2} < 1$, whose spectral radius is $1 + u < 2$. In both cases $S_\infty(s) \ll g^s\rho^s$ with $\rho < g$, which is the claim with $c = 1 - \log_g \rho$. □

Example 5.5. For $g = 2$, $f(\theta) = |\cos \pi\theta|$ and $\phi(x) = -\log\cos \pi x$. From $2x + y \geq \frac{1}{4}$ we get $\max(x, y) \geq \frac{1}{12}$, so $\chi(\frac{1}{4}) \geq -\log\cos(\pi/12) > 0.0346$, $u < e^{-0.0173}$, $1 + u < 1.9829$, and the proof gives $c_2 > 0.012$.

5.3. **Windows.** Lemma 5.3 uses Lemma 5.1 only as an inequality for each pair of neighbouring factors, and then halves. Lemma 5.2 allows us to do much better: we bound the supremum over all real starting phases on windows of consecutive factors, along the exact recursion. Transfer matrices with certified Perron–Frobenius bounds are a standard device for products over digits with a single frequency; the point here is that the supremum over the second frequency a can be taken window by window.

Fix a window length $w \geq 2$ and weights $\lambda_0, \dots, \lambda_{w-1} \geq 0$ with $K = \sum_j \lambda_j > 0$. For real $D_0, \dots, D_{w-2}$ put

$$\mathcal{U}(D_0, \dots, D_{w-2}) = \sup_{\theta\in\mathbb{R}} \prod_{j=0}^{w-1} f(\theta_j)^{\lambda_j}, \qquad \theta_0 = \theta, \quad \theta_{j+1} = g\theta_j - D_j.$$

Lemma 5.6. *If $w \leq s$, then for every ℓ, with D_i as in Lemma 5.2,*

$$\max_a |\Psi_\ell(a)| \leq g^s \prod_{i=0}^{s-w} \mathcal{U}(D_i, \dots, D_{i+w-2})^{1/K}.$$

Proof. Fix a. By Lemma 5.2 and the periodicity of f, for each $0 \leq i \leq s - w$ the numbers $\theta_i, \dots, \theta_{i+w-1}$ give the value of a product as in the definition of $\mathcal{U}(D_i, \dots, D_{i+w-2})$, so $\sum_j \lambda_j \log f(\theta_{i+j}) \leq \log\mathcal{U}(D_i, \dots, D_{i+w-2})$. Sum over i. Each index $m \in [0, s)$ receives total weight $\mathrm{cov}_m \leq K$, and $\log f \leq 0$, so $K\sum_m \log f(\theta_m) \leq \sum_m \mathrm{cov}_m \log f(\theta_m) \leq \sum_i \log\mathcal{U}(\cdots)$. Finally use Lemma 2.1(ii). □

Proposition 5.7. *Fix w and λ as above and an integer $r \geq 1$, and put $\nu = w + r - 2$; assume that $\nu \geq 2$. For $\sigma = (\sigma_0, \dots, \sigma_{\nu-1}) \in \{0, \dots, g-1\}^\nu$ let $v(\sigma) = \sum_{j=0}^{r-1} \sigma_{r-1-j} g^{-j-1}$ and*

$$\widehat{U}(\sigma) = \sup\Big\{\mathcal{U}(D_0, \dots, D_{w-2}) :\ D_0 \in \big[v(\sigma), v(\sigma) + g^{-r}\big],\ D_m = \frac{\sigma_{r-1+m} + D_{m-1}}{g}\Big\},$$

where $1 \leq m \leq w-2$. Let $\overline{U}(\sigma) \geq \widehat{U}(\sigma)$ for all σ, and let M be the non-negative matrix indexed by $\{0, \dots, g-1\}^{\nu-1}$ with

$$M\big[(\sigma_0, \dots, \sigma_{\nu-2}), (\sigma_1, \dots, \sigma_{\nu-1})\big] = \overline{U}(\sigma)^{1/K}$$

and all other entries 0. If $\mathbf{x}$ is a vector with positive entries and $\rho_{\mathbf{x}} = \max_i (M\mathbf{x})_i/\mathbf{x}_i$, then $S_\infty(s) \ll g^s \rho_{\mathbf{x}}^s$, where the implied constant depends on $g, w, r, \lambda, \overline{U}$ and $\mathbf{x}$. That is, Theorem 1.2 holds with $c_g = 1 - \log_g \rho_{\mathbf{x}}$.

Proof. For $s < w$ the claim follows from the trivial bound $S_\infty(s) \leq 2g^{2s}$, so let $s \geq w$. For the window starting at i put $\sigma^{(i)} = (\ell'_{i+2-r}, \dots, \ell'_{i+w-1})$, with $\ell'_j = 0$ for $j < 0$. By Lemma 5.2, the number D_i lies in the interval $[v(\sigma^{(i)}), v(\sigma^{(i)}) + g^{-r}]$, and $D_{i+m} = (\ell'_{i+m+1} + D_{i+m-1})/g$; hence $\mathcal{U}(D_i, \dots, D_{i+w-2}) \leq \overline{U}(\sigma^{(i)})$. As in the proof of Proposition 5.4, Lemmas 5.6 and 5.2 give

$$S_\infty(s) \leq 2g^s \sum_{\ell' \bmod g^s} \prod_{i=0}^{s-w} \overline{U}(\sigma^{(i)})^{1/K}.$$

Consecutive $\sigma^{(i)}$ overlap in $\nu-1$ digits, so each product is the weight of a path with $n = s-w+1$ edges in the graph of M. Summing also over the (fictitious) digits ℓ'_j, $j < 0$, and over ℓ'_0 when $r = 1$ (it then lies in no window), we see that the sum over ℓ' is at most $g\,\mathbf{1}^{\mathsf{T}} M^n \mathbf{1} \leq g^\nu \max_i (M^n\mathbf{1})_i$,

where $\mathbf{1}$ is the all-ones vector. Finally, $M\mathbf{x} \le \rho_{\mathbf{x}}\mathbf{x}$ entrywise and $M \ge 0$ give $M^n\mathbf{x} \le \rho_{\mathbf{x}}^n\mathbf{x}$, hence $M^n\mathbf{1} \le (\max_i \mathbf{x}_i / \min_i \mathbf{x}_i)\, \rho_{\mathbf{x}}^n \mathbf{1}$. □

The weights $\overline{U}(\sigma)^{1/K}$ are positive, since $\widehat{U}(\sigma) > 0$ (f vanishes only at finitely many points modulo 1), and the graph of M is a de Bruijn graph, which is strongly connected. So M is irreducible, and by the Collatz–Wielandt formula the infimum of $\rho_{\mathbf{x}}$ over all positive $\mathbf{x}$ is the spectral radius of M [8, Chapter 8]. In practice, power iteration produces a vector $\mathbf{x}$ for which $\rho_{\mathbf{x}}$ is very close to the spectral radius.

5.4. **Computing the constants.** We now describe how the values in Table 1 were obtained. A Python script that carries out the computation accompanies this paper (see Section 7.1).

Upper bounds for $\widehat{U}$. Parametrise an orbit by its last point. Put $C_0 = f^{\lambda_0}$ and

$$C_{m+1}(y) = f(y)^{\lambda_{m+1}} \max_{0 \le q < g} \sup_{D_m} C_m\Big(\frac{y + D_m + q}{g}\Big) \qquad (0 \le m \le w-2),$$

where D_m runs over the interval $[v_m, v_m + g^{-r-m}]$, with $v_0 = v(\sigma)$ and $v_m = (\sigma_{r-1+m} + v_{m-1})/g$. Then $\widehat{U}(\sigma) \le \sup_y C_{w-1}(y)$. Indeed, the numbers $(y + D_m + q)/g$, $0 \le q < g$, are the preimages of y modulo 1 under $\theta \mapsto g\theta - D_m$, so by induction $C_m(y)$ is at least $\prod_{j \le m} f(\theta_j)^{\lambda_j}$ for every orbit with $\theta_m \equiv y \pmod 1$; treating the D_m as independent in their intervals only enlarges the bound.

Since the argument of C_m is contracted by g, a fixed grid suffices. We use the closed cells $[n/N, (n+1)/N]$, $0 \le n < N$, and the following upper bound for f on each cell. It is 1 on the two cells that contain an integer. On the other cells, $|\sin \pi\theta|$ is minimal at an endpoint, and $|\sin \pi g\theta|$ is at most 1, and is maximal at an endpoint if the cell contains no θ with $g\theta \in \frac{1}{2} + \mathbb{Z}$. These bounds are propagated through the recursion: the bound for C_{m+1} on a cell is the bound for $f^{\lambda_{m+1}}$ on that cell times the maximum of the bounds for C_m over blocks of consecutive cells that contain the preimages of the cell, for all D_m in its interval. These blocks are computed in exact integer arithmetic, and no Lipschitz constants are needed.

Rounding. The cell bounds for f are evaluated in 40-digit decimal arithmetic, multiplied by $1 + 10^{-12}$, and rounded upward to double precision. All later steps are carried out in double precision, where maxima are exact; we assume, as holds for standard implementations, that each power, product, sum or quotient has relative error at most 2^{-52} (any bound below 10^{-14} would do). The script also checks that every computed $\overline{U}(\sigma)$ exceeds 10^{-280}, so underflow plays no role. Consider one step of the recursion on one cell. If the cell bound for f equals 1, the step consists of maxima and a multiplication by 1, which are exact. Otherwise the cell bound for f exceeds the exact one by a factor of about $1 + 10^{-12}$; since $\lambda_{m+1} \ge 1$ (all our weights are positive integers), this factor covers the rounding errors of the power and of the product. By induction, the computed bounds for C_m are at least the exact ones, so the computed values $\overline{U}(\sigma)$ are upper bounds for $\widehat{U}(\sigma)$. In the last stage, each ratio $(M\mathbf{x})_i/\mathbf{x}_i$ involves $3g$ roundings (g powers, g products, $g-1$ sums and one quotient) and the rounding of the exponent $1/K$, whose relative effect on an entry e is at most $|\log e|\, 2^{-53} < 10^{-13}/2$, because $e > 10^{-140}$. The total relative effect is below 10^{-13}, and we multiply the final maximum by $1 + 10^{-12}$.

Parameters. We use triangular weights $\lambda_j = \min(j+1, w-j)$, which reduce the edge effects of the windows, $N = 8192$, and the vector $\mathbf{x}$ produced by power iteration. Table 2 lists the parameters and the resulting upper bounds ρ_{CW} for $\rho_{\mathbf{x}}$; then $c_g = 1 - \log_g \rho_{\mathrm{CW}}$, rounded down, is the value in Table 1. The cost grows like $g^{w+r-1}N$, which limits w for large g.

Proof of Theorem 1.2. The existence of $c_g > 0$ for every g is Proposition 5.4. The values in Table 1 follow from Proposition 5.7 and the computation just described. □

TABLE 2. Window parameters, certified bounds ρ_{CW} (rounded up) and the resulting values of c_g (rounded down). All rows use triangular weights and $N = 8192$.

g	w	r	ρ_{CW}	c_g
2	14	5	1.832340	0.1263
3	10	3	2.687053	0.1002
4	7	3	3.574206	0.0811
5	6	3	4.460721	0.0709
6	6	2	5.348980	0.0641
7	6	2	6.231057	0.0597
8	5	2	7.145309	0.0543
9	5	2	8.034354	0.0516
10	6	1	8.919963	0.0496
12	5	2	10.706920	0.0458
16	4	2	14.387430	0.0383

Remark 5.8. For $g = 2$ and $r = 5$, uniform weights give $c_2 \geq 0.114$, 0.117, 0.119 for $w = 10, 12, 14$, and triangular weights give 0.121, 0.124, 0.126. Numerically, $S_\infty(s+1)/S_\infty(s)$ settles at about $2^{1.866}$ ($s \leq 16$), which corresponds to $c_2 \approx 0.134$; for $g = 3$ and $g = 4$ the corresponding values are about 0.108 and 0.090, and for $g = 10$ the ratio at $s = 4$ corresponds to $c_{10} \approx 0.064$. If these values reflect the true growth of S_∞, then no bound for S_∞ can improve γ much further for $g = 2$: with $c_2 = 0.134$, Theorem 1.1(a) would give $\gamma \approx 0.965$, against our 0.9674. The remaining gap comes partly from the window length and partly from the discretisation. The pairing bound of Proposition 5.4, with more digits r and the same discretisation, gives only $c_2 \approx 0.06$ and $c_{10} \approx 0.03$.

Remark 5.9. The power saving for S_∞ uses the special structure of digit reversal only through the product formula of Lemma 2.1(ii) and the identity of Lemma 5.1. For a general permutation π of $[0, g^s)$, let $\Psi_\ell^\pi(a)$ be defined as $\Psi_\ell(a)$ with π in place of rev_s. Then $\max_a |\Psi_\ell^\pi(a)|$ can equal g^s for every ℓ, so that the analogue of $S_\infty(s)$ is of order g^{2s}; for instance, $\pi(m) = hm \bmod g^s$ with $\gcd(h, g) = 1$ gives $|\Psi_\ell^\pi(-h\ell)| = g^s$.

6. Proof of Theorem 1.1

Remark 6.1. For a prime power g, Lemma 4.2 applies to every node with $D = g^{1/k}$, a constant. If g is not a prime power, a node can be divisible by a high power of one prime factor of g and still satisfy $g \nmid b^k$ (in base 10, for example $b = 2^v$). For such nodes both d_b and the multiplicity of $b \mapsto T_b$ are about $m_g(b)^{k-1}$, and the corresponding progressions resonate together. We therefore treat the integers x with large g-part separately, by a direct count.

Lemma 6.2 (large g-part)**.** *Let P be the largest prime factor of g, and let $D \geq 1$ with $PD \leq X_1^{\log 2/(2\log g)}$. Then*

$$\#\{x : \ x^k \in \mathcal{P}_{g,L}, \ m_g(x) > D\} \ \ll \ \tau_g(PD)\, X_1^{1/2} D^{-1}.$$

Proof. Multiplying in the prime factors of $m_g(x)$ one at a time, we find a divisor m' of $m_g(x)$ with $D < m' \leq PD$. Fix such an m', and choose t' with $X_1^{1/2} \leq g^{t'} < gX_1^{1/2}$. Then $p^{v_p(m')} \leq PD \leq 2^{t'} \leq p^{v_p(g)t'}$ for each $p \mid g$, so $m' \mid g^{t'}$. Apply Lemma 3.1 with t' in place of t (this part of Lemma 3.1 does not use s): the x in question with $m' \mid x$ lie in nodes b of level t' with $m' \mid b$, of which there are at most $g^{t'}/m'$. By the first assertion of Lemma 3.2 (which does not involve

s, and where now $X_1 g^{-2t'} \le 1$), each such node contains at most $g/k+1$ of them. So there are $\ll X_1^{1/2}/m' \le X_1^{1/2}/D$ such x, and there are at most $\tau_g(PD)$ choices of m'. □

Proof of Theorem 1.1. Recall $X_1 \asymp (\#\mathcal{P}_{g,L})^{2/k}$, and note $\frac{2}{k}\cdot\frac{2-c}{4-c} = \frac{1}{k}\big(1-\frac{c}{4-c}\big)$. We may assume that L is large.

(a) Let $g = p^e$, $D = g^{1/k}$, and let $\mathcal{B}$ be the set of all nodes; by Lemma 4.2 every node has $m_g(b) < D$, and $\tau \ll 1$. Choose t with $g^{(4-c)t} \asymp X_1$ and put $s = t-2$. Then $Z = X_1 g^{-2t} \asymp g^{(2-c)t}$; for large L we have $2k \le Z$, $D^k = g < 2^t$, $D^{k-1}Z \le g^{2s}$, and $t+s < 2t \le L/2$ because $t = L/(k(4-c)) + O(1)$ and $k(4-c) > 6$. By Lemma 3.1 every solution x satisfies $x \bmod g^t \in \mathcal{B}$. By Proposition 3.4, Lemma 3.2 and Corollary 4.4, using $g^{t-s} = g^2$ and $s \asymp \log Z \asymp \log X_1$,

$$\#\mathcal{Q}_{g,L}^k \ll Z + 1 + (\log X_1)\big(S_\infty(s) + Z\big) \ll (\log X_1)\, X_1^{\frac{2-c}{4-c}},$$

since $S_\infty(s) \ll g^{(2-c)s} \le g^{(2-c)t} \asymp Z \asymp X_1^{(2-c)/(4-c)}$.

(b) Let

$$\eta = \frac{c}{2\big((k-1)(4-c)+2\big)}, \qquad D = X_1^\eta.$$

Split the x with $x^k \in \mathcal{Q}_{g,L}^k$ according to whether $m_g(x) > D$. Those with $m_g(x) > D$ number $\ll (\log X_1)^{\omega(g)} X_1^{1/2-\eta}$ by Lemma 6.2; its condition holds because $\eta < c/4 \le \log 2/(8\log g)$.

For the others, choose t with $g^{(4-c)t} \asymp X_1/D$, put $s = t-2$, and let $\mathcal{B}$ be the set of nodes b with $m_g(b) \le D$. Then $t = (1-\eta)L/(k(4-c)) + O(1)$, so for large L we have $t \ge 3$ and $t+s < 2t \le L/2$. Moreover $D^k < 2^t$ for large L, because

$$k\eta \le \frac{c}{4-c} \le \frac{\log 2}{2(4-c)\log g} < (1-\eta)\frac{\log 2}{(4-c)\log g} \quad\text{and}\quad 2^t \gg X_1^{(1-\eta)\log 2/((4-c)\log g)}.$$

If $m_g(x) \le D$ then $p^{v_p(x)} \le D < 2^t \le p^{v_p(g)t}$ for each $p \mid g$, so $m_g(x \bmod g^t) = m_g(x)$, and by Lemma 3.1, $x \bmod g^t \in \mathcal{B}$. Now $Z = X_1 g^{-2t} \asymp X_1^{\frac{2-c}{4-c}} D^{\frac{2}{4-c}}$ and $Dg^{(2-c)t} \asymp X_1^{\frac{2-c}{4-c}} D^{\frac{2}{4-c}}$ as well; in particular $Z \ge 2k$ for large L, so Lemma 4.3 applies. We shall also use $D^{k-1}Z \le g^{2s}$, which holds for large L because $\eta\big((k-1)(4-c)+4\big) < c$. By Proposition 3.4, Lemma 3.2 and Corollary 4.4, with $\tau \ll (\log X_1)^{\omega(g)}$ and $g^{t-s} = g^2$,

$$\begin{aligned}\#\{x:\ m_g(x) \le D\} &\ll Z + \tau(\log X_1)\Big(D^{k-1}S_\infty(s) + D^{2k-2}Zg^{-2s}S_\infty(s) + D^{k-2}Z\Big)\\ &\ll (\log X_1)^{\omega(g)+1} D^{k-2}\big(D\,g^{(2-c)t} + Z\big),\end{aligned}$$

using $D^{k-1}Z \le g^{2s}$ in the middle term. The right side is $\ll (\log X_1)^{\omega(g)+1} D^{k-2+\frac{2}{4-c}} X_1^{\frac{2-c}{4-c}}$. With $D = X_1^\eta$ the exponent of X_1 is

$$\eta\Big(k-2+\frac{2}{4-c}\Big) + \frac{2-c}{4-c} = \frac{1}{2} - \eta,$$

by the choice of η. Finally $2(\frac12 - \eta) = 1 - \frac{c}{(k-1)(4-c)+2}$. □

Proof of Corollary 1.3. The first statement follows from Theorems 1.1 and 1.2, replacing c by $\min(c_g, \frac{\log 2}{2\log g})$ in part (b). For the table, apply Theorem 1.1(a) for prime powers and (b) for $g = 6, 10, 12$, with $c = c_g$ from Table 1; these satisfy $c_g \le \log 2/(2\log g)$. The table entries are the resulting exponents rounded up in the fourth decimal, which absorbs the logarithmic factors. □

7. Further remarks

7.1. **Numerical checks.** The script `window_constants.py`, which accompanies this paper, implements Section 5.4: for each base it prints ρ_{CW} and c_g, and it reproduces Table 2 in about an hour on one processor core. In a second mode it computes, for small s, the values of $\max_a |\Psi_\ell(a)|$ by the fast Fourier transform (in floating point), and compares them for every ℓ with the bound of Lemma 5.6, with $\overline{U}$ in place of $\mathcal{U}$. For $(g,s,w,r) = (2,12,8,5)$, $(3,7,5,3)$, $(4,7,5,3)$, $(6,5,4,2)$, $(10,4,3,1)$, $(10,4,3,2)$ and $(16,3,3,1)$, with $N = 4096$, the bound held for every ℓ; the largest ratio was 0.99995, for $g = 16$. Finally, we checked the identity of Lemma 5.1 exhaustively for small (g,s), Lemma 5.2 in exact rational arithmetic, and the structural statements of Lemma 4.2 for all nodes for $(g,k,t) = (10,2,4)$, $(10,2,5)$, $(10,3,4)$, $(6,2,5)$, $(6,3,5)$ and $(12,2,4)$; these checks are not needed for the proofs.

7.2. **Limits of the method.** The argument bounds the weights $|\Psi_\ell(a)|$ by their maximum over a and uses averaging only over the archimedean phases. By Remark 5.8, this is unlikely to give much more than $\gamma \approx 0.965$ in base 2. Going further would require cancellation in the sum over a, that is, information on the joint distribution of the archimedean phases $\ell\delta_b$ and the g-adic phases $a\beta_b$ across the nodes. For bases that are not prime powers, the treatment of integers with a large g-part (Lemma 6.2) is also wasteful and could probably be improved.

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